

MODULE 3

Statistical Experiments and Significance Testing

Design of experiments is a cornerstone of the practice of statistics, with applications in virtually all areas of research. The goal is to design an experiment in order to confirm or reject a hypothesis. Data scientists often need to conduct continual experiments, particularly regarding user interface and product marketing. This chapter reviews traditional experimental design and discusses some common challenges in data science. It also covers some oft-cited concepts in statistical inference and explains their meaning and relevance (or lack of relevance) to data science.

Whenever you see references to statistical significance, t-tests, or p-values, it is typically in the context of the classical statistical inference “pipeline” (see [Figure 3-1](#)). This process starts with a hypothesis (“drug A is better than the existing standard drug,” or “price A is more profitable than the existing price B”). An experiment (it might be an A/B test) is designed to test the hypothesis—designed in such a way that it hopefully will deliver conclusive results. The data is collected and analyzed, and then a conclusion is drawn. The term *inference* reflects the intention to apply the experiment results, which involve a limited set of data, to a larger process or population.

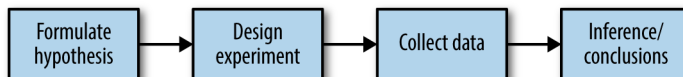


Figure 3-1. The classical statistical inference pipeline

A/B Testing

An A/B test is an experiment with two groups to establish which of two treatments, products, procedures, or the like is superior. Often one of the two treatments is the standard existing treatment, or no treatment. If a standard (or no) treatment is used, it is called the *control*. A typical hypothesis is that a new treatment is better than the control.

Key Terms for A/B Testing

Treatment

Something (drug, price, web headline) to which a subject is exposed.

Treatment group

A group of subjects exposed to a specific treatment.

Control group

A group of subjects exposed to no (or standard) treatment.

Randomization

The process of randomly assigning subjects to treatments.

Subjects

The items (web visitors, patients, etc.) that are exposed to treatments.

Test statistic

The metric used to measure the effect of the treatment.

A/B tests are common in web design and marketing, since results are so readily measured. Some examples of A/B testing include:

- Testing two soil treatments to determine which produces better seed germination
- Testing two therapies to determine which suppresses cancer more effectively
- Testing two prices to determine which yields more net profit
- Testing two web headlines to determine which produces more clicks (Figure 3-2)
- Testing two web ads to determine which generates more conversions

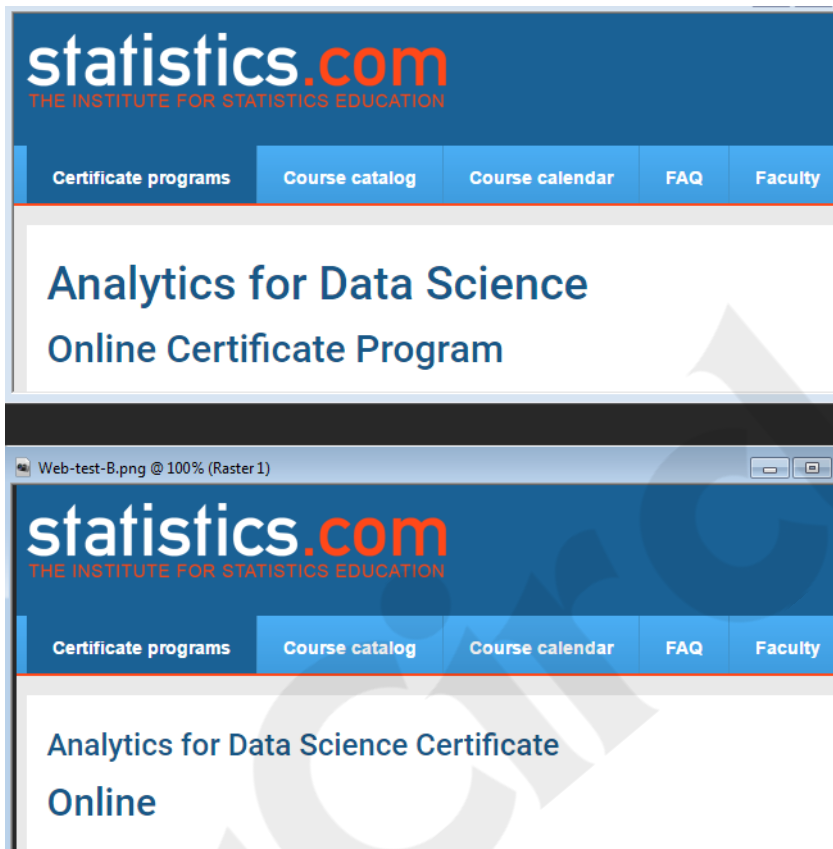


Figure 3-2. Marketers continually test one web presentation against another

A proper A/B test has *subjects* that can be assigned to one treatment or another. The subject might be a person, a plant seed, a web visitor; the key is that the subject is exposed to the treatment. Ideally, subjects are *randomized* (assigned randomly) to treatments. In this way, you know that any difference between the treatment groups is due to one of two things:

- The effect of the different treatments
- Luck of the draw in which subjects are assigned to which treatments (i.e., the random assignment may have resulted in the naturally better-performing subjects being concentrated in A or B)

You also need to pay attention to the *test statistic* or metric you use to compare group A to group B. Perhaps the most common metric in data science is a binary variable: click or no-click, buy or don't buy, fraud or no fraud, and so on. Those results would be summed up in a 2×2 table. [Table 3-1](#) is a 2×2 table for an actual price test (see “Statistical Significance and p-Values” on [page 103](#) for further discussion of these results).

Table 3-1. 2×2 table for ecommerce experiment results

Outcome	Price A	Price B
Conversion	200	182
No conversion	23,539	22,406

If the metric is a continuous variable (purchase amount, profit, etc.) or a count (e.g., days in hospital, pages visited), the result might be displayed differently. If one were interested not in conversion but in revenue per page view, the results of the price test in [Table 3-1](#) might look like this in typical default software output:

Revenue/page view with price A: mean = 3.87, SD = 51.10

Revenue/page view with price B: mean = 4.11, SD = 62.98

“SD” refers to the standard deviation of the values within each group.



Just because statistical software—including *R* and *Python*—generates output by default does not mean that all the output is useful or relevant. You can see that the preceding standard deviations are not that useful; on their face they suggest that numerous values might be negative, when negative revenue is not feasible. This data consists of a small set of relatively high values (page views with conversions) and a huge number of 0-values (page views with no conversion). It is difficult to sum up the variability of such data with a single number, though the mean absolute deviation from the mean (7.68 for A and 8.15 for B) is more reasonable than the standard deviation.

Why Have a Control Group?

Why not skip the control group and just run an experiment applying the treatment of interest to only one group, and compare the outcome to prior experience?

Without a control group, there is no assurance that “all other things are equal” and that any difference is really due to the treatment (or to chance). When you have a control group, it is subject to the same conditions (except for the treatment of

interest) as the treatment group. If you simply make a comparison to “baseline” or prior experience, other factors, besides the treatment, might differ.



Blinding in studies

A *blind study* is one in which the subjects are unaware of whether they are getting treatment A or treatment B. Awareness of receiving a particular treatment can affect response. A *double-blind* study is one in which the investigators and facilitators (e.g., doctors and nurses in a medical study) also are unaware which subjects are getting which treatment. Blinding is not possible when the nature of the treatment is transparent—for example, cognitive therapy from a computer versus a psychologist.

A/B testing in data science is typically used in a web context. Treatments might be the design of a web page, the price of a product, the wording of a headline, or some other item. Some thought is required to preserve the principles of randomization. Typically the subject in the experiment is the web visitor, and the outcomes we are interested in measuring are clicks, purchases, visit duration, number of pages visited, whether a particular page is visited, and the like. In a standard A/B experiment, you need to decide on one metric ahead of time. Multiple behavior metrics might be collected and be of interest, but if the experiment is expected to lead to a decision between treatment A and treatment B, a single metric, or *test statistic*, needs to be established beforehand. Selecting a test statistic *after* the experiment is conducted opens the door to researcher bias.

Why Just A/B? Why Not C, D,...?

A/B tests are popular in the marketing and ecommerce worlds, but are far from the only type of statistical experiment. Additional treatments can be included. Subjects might have repeated measurements taken. Pharmaceutical trials where subjects are scarce, expensive, and acquired over time are sometimes designed with multiple opportunities to stop the experiment and reach a conclusion.

Traditional statistical experimental designs focus on answering a static question about the efficacy of specified treatments. Data scientists are less interested in the question:

Is the difference between price A and price B statistically significant?

than in the question:

Which, out of multiple possible prices, is best?

For this, a relatively new type of experimental design is used: the *multi-arm bandit* (see “[Multi-Arm Bandit Algorithm](#)” on page 131).



Getting Permission

In scientific and medical research involving human subjects, it is typically necessary to get their permission, as well as obtain the approval of an institutional review board. Experiments in business that are done as a part of ongoing operations almost never do this. In most cases (e.g., pricing experiments, or experiments about which headline to show or which offer should be made), this practice is widely accepted. Facebook, however, ran afoul of this general acceptance in 2014 when it experimented with the emotional tone in users' newsfeeds. Facebook used sentiment analysis to classify newsfeed posts as positive or negative, and then altered the positive/negative balance in what it showed users. Some randomly selected users experienced more positive posts, while others experienced more negative posts. Facebook found that the users who experienced a more positive newsfeed were more likely to post positively themselves, and vice versa. The magnitude of the effect was small, however, and Facebook faced much criticism for conducting the experiment without users' knowledge. Some users speculated that Facebook might have pushed some extremely depressed users over the edge if they got the negative version of their feed.

Key Ideas

- Subjects are assigned to two (or more) groups that are treated exactly alike, except that the treatment under study differs from one group to another.
- Ideally, subjects are assigned randomly to the groups.

Further Reading

- Two-group comparisons (A/B tests) are a staple of traditional statistics, and just about any introductory statistics text will have extensive coverage of design principles and inference procedures. For a discussion that places A/B tests in more of a data science context and uses resampling, see *Introductory Statistics and Analytics: A Resampling Perspective* by Peter Bruce (Wiley, 2014).
- For web testing, the logistical aspects of testing can be just as challenging as the statistical ones. A good place to start is the [Google Analytics help section on experiments](#).
- Beware advice found in the ubiquitous guides to A/B testing that you see on the web, such as these words in one such guide: "Wait for about 1,000 total visitors and make sure you run the test for a week." Such general rules of thumb are not

statistically meaningful; see “Power and Sample Size” on page 135 for more detail.

Hypothesis Tests

Hypothesis tests, also called *significance tests*, are ubiquitous in the traditional statistical analysis of published research. Their purpose is to help you learn whether random chance might be responsible for an observed effect.

Key Terms for Hypothesis Tests

Null hypothesis

The hypothesis that chance is to blame.

Alternative hypothesis

Counterpoint to the null (what you hope to prove).

One-way test

Hypothesis test that counts chance results only in one direction.

Two-way test

Hypothesis test that counts chance results in two directions.

An A/B test (see “A/B Testing” on page 88) is typically constructed with a hypothesis in mind. For example, the hypothesis might be that price B produces higher profit. Why do we need a hypothesis? Why not just look at the outcome of the experiment and go with whichever treatment does better?

The answer lies in the tendency of the human mind to underestimate the scope of natural random behavior. One manifestation of this is the failure to anticipate extreme events, or so-called “black swans” (see “Long-Tailed Distributions” on page 73). Another manifestation is the tendency to misinterpret random events as having patterns of some significance. Statistical hypothesis testing was invented as a way to protect researchers from being fooled by random chance.

Misinterpreting Randomness

You can observe the human tendency to underestimate randomness in this experiment. Ask several friends to invent a series of 50 coin flips: have them write down a series of random Hs and Ts. Then ask them to actually flip a coin 50 times and write down the results. Have them put the real coin flip results in one pile, and the made-up results in another. It is easy to tell which results are real: the real ones will have longer runs of Hs or Ts. In a set of 50 *real* coin flips, it is not at all unusual to see five

or six Hs or Ts in a row. However, when most of us are inventing random coin flips and we have gotten three or four Hs in a row, we tell ourselves that, for the series to look random, we had better switch to T.

The other side of this coin, so to speak, is that when we *do* see the real-world equivalent of six Hs in a row (e.g., when one headline outperforms another by 10%), we are inclined to attribute it to something real, not just to chance.

In a properly designed A/B test, you collect data on treatments A and B in such a way that any observed difference between A and B must be due to either:

- Random chance in assignment of subjects
- A true difference between A and B

A statistical hypothesis test is further analysis of an A/B test, or any randomized experiment, to assess whether random chance is a reasonable explanation for the observed difference between groups A and B.

The Null Hypothesis

Hypothesis tests use the following logic: “Given the human tendency to react to unusual but random behavior and interpret it as something meaningful and real, in our experiments we will require proof that the difference between groups is more extreme than what chance might reasonably produce.” This involves a baseline assumption that the treatments are equivalent, and any difference between the groups is due to chance. This baseline assumption is termed the *null hypothesis*. Our hope, then, is that we can in fact prove the null hypothesis *wrong* and show that the outcomes for groups A and B are more different than what chance might produce.

One way to do this is via a resampling permutation procedure, in which we shuffle together the results from groups A and B and then repeatedly deal out the data in groups of similar sizes, and then observe how often we get a difference as extreme as the observed difference. The combined shuffled results from groups A and B, and the procedure of resampling from them, embodies the null hypothesis of groups A and B being equivalent and interchangeable and is termed the null model. See “[Resampling](#)” on page 96 for more detail.

Alternative Hypothesis

Hypothesis tests by their nature involve not just a null hypothesis but also an offsetting alternative hypothesis. Here are some examples:

- Null = “no difference between the means of group A and group B”; alternative = “A is different from B” (could be bigger or smaller)
- Null = “ $A \leq B$ ”; alternative = “ $A > B$ ”
- Null = “B is not X% greater than A”; alternative = “B is X% greater than A”

Taken together, the null and alternative hypotheses must account for all possibilities. The nature of the null hypothesis determines the structure of the hypothesis test.

One-Way Versus Two-Way Hypothesis Tests

Often in an A/B test, you are testing a new option (say, B) against an established default option (A), and the presumption is that you will stick with the default option unless the new option proves itself definitively better. In such a case, you want a hypothesis test to protect you from being fooled by chance in the direction favoring B. You don’t care about being fooled by chance in the other direction, because you would be sticking with A unless B proves definitively better. So you want a *directional* alternative hypothesis (B is better than A). In such a case, you use a *one-way* (or one-tail) hypothesis test. This means that extreme chance results in only one direction count toward the p-value.

If you want a hypothesis test to protect you from being fooled by chance in either direction, the alternative hypothesis is *bidirectional* (A is different from B; could be bigger or smaller). In such a case, you use a *two-way* (or two-tail) hypothesis. This means that extreme chance results in either direction count toward the p-value.

A one-tail hypothesis test often fits the nature of A/B decision making, in which a decision is required and one option is typically assigned “default” status unless the other proves better. Software, however, including *R* and *scipy* in *Python*, typically provides a two-tail test in its default output, and many statisticians opt for the more conservative two-tail test just to avoid argument. One-tail versus two-tail is a confusing subject, and not that relevant for data science, where the precision of p-value calculations is not terribly important.

Key Ideas

- A null hypothesis is a logical construct embodying the notion that nothing special has happened, and any effect you observe is due to random chance.
- The hypothesis test assumes that the null hypothesis is true, creates a “null model” (a probability model), and tests whether the effect you observe is a reasonable outcome of that model.

Further Reading

- *The Drunkard's Walk* by Leonard Mlodinow (Pantheon, 2008) is a readable survey of the ways in which “randomness rules our lives.”
- David Freedman, Robert Pisani, and Roger Purves’s classic statistics text *Statistics*, 4th ed. (W. W. Norton, 2007), has excellent nonmathematical treatments of most statistics topics, including hypothesis testing.
- *Introductory Statistics and Analytics: A Resampling Perspective* by Peter Bruce (Wiley, 2014) develops hypothesis testing concepts using resampling.

Resampling

Resampling in statistics means to repeatedly sample values from observed data, with a general goal of assessing random variability in a statistic. It can also be used to assess and improve the accuracy of some machine-learning models (e.g., the predictions from decision tree models built on multiple bootstrapped data sets can be averaged in a process known as *bagging*—see “[Bagging and the Random Forest](#)” on page 259).

There are two main types of resampling procedures: the *bootstrap* and *permutation* tests. The bootstrap is used to assess the reliability of an estimate; it was discussed in the previous chapter (see “[The Bootstrap](#)” on page 61). Permutation tests are used to test hypotheses, typically involving two or more groups, and we discuss those in this section.

Key Terms for Resampling

Permutation test

The procedure of combining two or more samples together and randomly (or exhaustively) reallocating the observations to resamples.

Synonyms

Randomization test, random permutation test, exact test

Resampling

Drawing additional samples (“resamples”) from an observed data set.

With or without replacement

In sampling, whether or not an item is returned to the sample before the next draw.

Permutation Test

In a *permutation* procedure, two or more samples are involved, typically the groups in an A/B or other hypothesis test. *Permute* means to change the order of a set of values. The first step in a *permutation test* of a hypothesis is to combine the results from groups A and B (and, if used, C, D,...). This is the logical embodiment of the null hypothesis that the treatments to which the groups were exposed do not differ. We then test that hypothesis by randomly drawing groups from this combined set and seeing how much they differ from one another. The permutation procedure is as follows:

1. Combine the results from the different groups into a single data set.
2. Shuffle the combined data and then randomly draw (without replacement) a resample of the same size as group A (clearly it will contain some data from the other groups).
3. From the remaining data, randomly draw (without replacement) a resample of the same size as group B.
4. Do the same for groups C, D, and so on. You have now collected one set of resamples that mirror the sizes of the original samples.
5. Whatever statistic or estimate was calculated for the original samples (e.g., difference in group proportions), calculate it now for the resamples, and record; this constitutes one permutation iteration.
6. Repeat the previous steps R times to yield a permutation distribution of the test statistic.

Now go back to the observed difference between groups and compare it to the set of permuted differences. If the observed difference lies well within the set of permuted differences, then we have not proven anything—the observed difference is within the range of what chance might produce. However, if the observed difference lies outside most of the permutation distribution, then we conclude that chance is *not* responsible. In technical terms, the difference is *statistically significant*. (See “Statistical Significance and p-Values” on page 103.)

Example: Web Stickiness

A company selling a relatively high-value service wants to test which of two web presentations does a better selling job. Due to the high value of the service being sold, sales are infrequent and the sales cycle is lengthy; it would take too long to accumulate enough sales to know which presentation is superior. So the company decides to measure the results with a proxy variable, using the detailed interior page that describes the service.



A *proxy* variable is one that stands in for the true variable of interest, which may be unavailable, too costly, or too time-consuming to measure. In climate research, for example, the oxygen content of ancient ice cores is used as a proxy for temperature. It is useful to have at least *some* data on the true variable of interest, so the strength of its association with the proxy can be assessed.

One potential proxy variable for our company is the number of clicks on the detailed landing page. A better one is how long people spend on the page. It is reasonable to think that a web presentation (page) that holds people’s attention longer will lead to more sales. Hence, our metric is average session time, comparing page A to page B.

Due to the fact that this is an interior, special-purpose page, it does not receive a huge number of visitors. Also note that Google Analytics, which is how we measure session time, cannot measure session time for the last session a person visits. Instead of deleting that session from the data, though, Google Analytics records it as a zero, so the data requires additional processing to remove those sessions. The result is a total of 36 sessions for the two different presentations, 21 for page A and 15 for page B. Using `ggplot`, we can visually compare the session times using side-by-side boxplots:

```
ggplot(session_times, aes(x=Page, y=Time)) +  
  geom_boxplot()
```

The pandas boxplot command uses the keyword argument `by` to create the figure:

```
ax = session_times.boxplot(by='Page', column='Time')
ax.set_xlabel('')
ax.set_ylabel('Time (in seconds)')
plt.suptitle('')
```

The boxplot, shown in [Figure 3-3](#), indicates that page B leads to longer sessions than page A. The means for each group can be computed in R as follows:

```
mean_a <- mean(session_times[session_times['Page'] == 'Page A', 'Time'])
mean_b <- mean(session_times[session_times['Page'] == 'Page B', 'Time'])
mean_b - mean_a
[1] 35.66667
```

In *Python*, we filter the pandas data frame first by page and then determine the mean of the `Time` column:

```
mean_a = session_times[session_times.Page == 'Page A'].Time.mean()
mean_b = session_times[session_times.Page == 'Page B'].Time.mean()
mean_b - mean_a
```

Page B has session times that are greater than those of page A by 35.67 seconds, on average. The question is whether this difference is within the range of what random chance might produce, i.e., is statistically significant. One way to answer this is to apply a permutation test—combine all the session times together and then repeatedly shuffle and divide them into groups of 21 (recall that $n_A = 21$ for page A) and 15 ($n_B = 15$ for page B).

To apply a permutation test, we need a function to randomly assign the 36 session times to a group of 21 (page A) and a group of 15 (page B). The R version of this function is:

```
perm_fun <- function(x, nA, nB)
{
  n <- nA + nB
  idx_b <- sample(1:n, nB)
  idx_a <- setdiff(1:n, idx_b)
  mean_diff <- mean(x[idx_b]) - mean(x[idx_a])
  return(mean_diff)
}
```

The *Python* version of this permutation test is the following:

```
def perm_fun(x, nA, nB):
    n = nA + nB
    idx_B = set(random.sample(range(n), nB))
    idx_A = set(range(n)) - idx_B
    return x.loc[idx_B].mean() - x.loc[idx_A].mean()
```

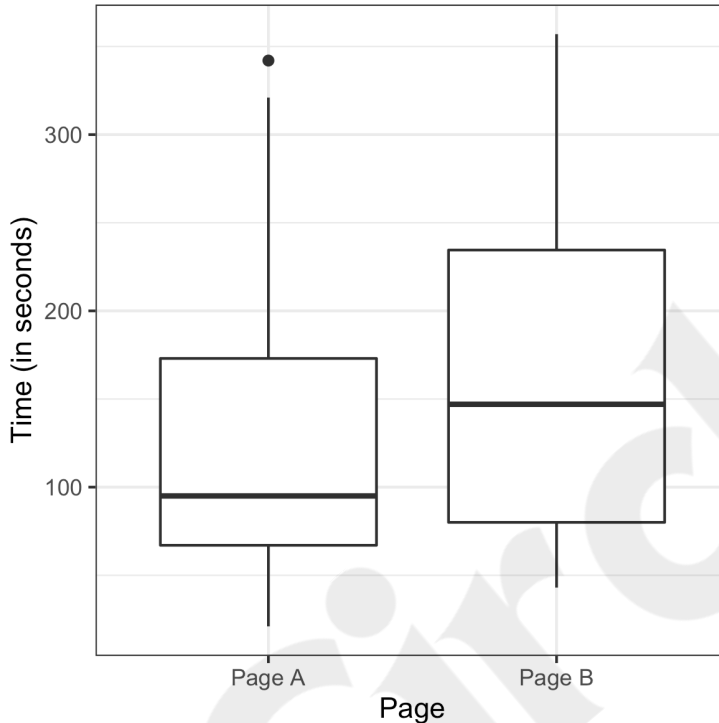


Figure 3-3. Session times for web pages A and B

This function works by sampling (without replacement) n_B indices and assigning them to the B group; the remaining n_A indices are assigned to group A. The difference between the two means is returned. Calling this function $R = 1,000$ times and specifying $n_A = 21$ and $n_B = 15$ leads to a distribution of differences in the session times that can be plotted as a histogram. In R this is done as follows using the `hist` function:

```
perm_diffs <- rep(0, 1000)
for (i in 1:1000) {
  perm_diffs[i] = perm_fun(session_times[, 'Time'], 21, 15)
}
hist(perm_diffs, xlab='Session time differences (in seconds)')
abline(v=mean_b - mean_a)
```

In *Python*, we can create a similar graph using `matplotlib`:

```
perm_diffs = [perm_fun(session_times.Time, nA, nB) for _ in range(1000)]

fig, ax = plt.subplots(figsize=(5, 5))
ax.hist(perm_diffs, bins=11, rwidth=0.9)
ax.axvline(x = mean_b - mean_a, color='black', lw=2)
```

```
ax.text(50, 190, 'Observed\ndifference', bbox={'facecolor':'white'})
ax.set_xlabel('Session time differences (in seconds)')
ax.set_ylabel('Frequency')
```

The histogram, in **Figure 3-4** shows that mean difference of random permutations often exceeds the observed difference in session times (the vertical line). For our results, this happens in 12.6% of the cases:

```
mean(perm_diffs > (mean_b - mean_a))
---
0.126
```

As the simulation uses random numbers, the percentage will vary. For example, in the *Python* version, we got 12.1%:

```
np.mean(perm_diffs > mean_b - mean_a)
---
0.121
```

This suggests that the observed difference in session time between page A and page B is well within the range of chance variation and thus is not statistically significant.

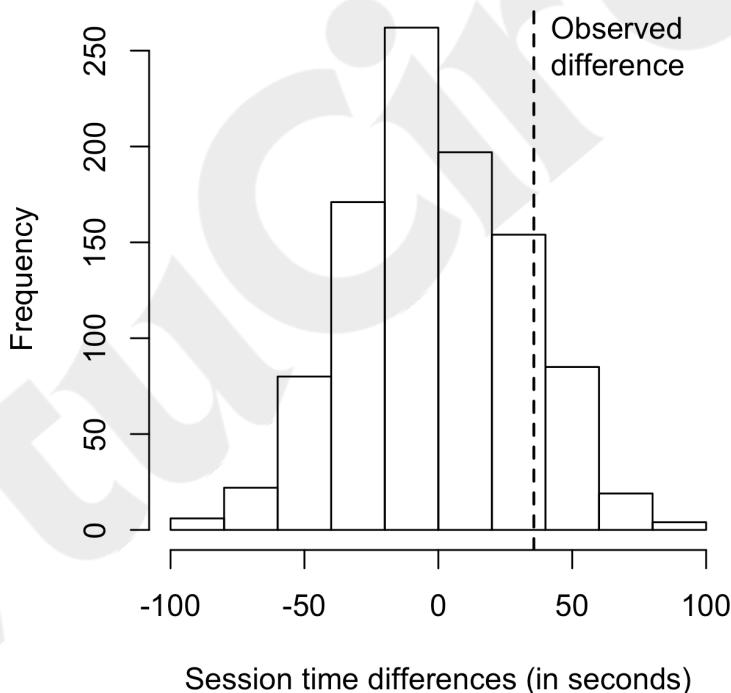


Figure 3-4. Frequency distribution for session time differences between pages A and B; the vertical line shows the observed difference

Exhaustive and Bootstrap Permutation Tests

In addition to the preceding random shuffling procedure, also called a *random permutation test* or a *randomization test*, there are two variants of the permutation test:

- An *exhaustive permutation test*
- A *bootstrap permutation test*

In an exhaustive permutation test, instead of just randomly shuffling and dividing the data, we actually figure out all the possible ways it could be divided. This is practical only for relatively small sample sizes. With a large number of repeated shufflings, the random permutation test results approximate those of the exhaustive permutation test, and approach them in the limit. Exhaustive permutation tests are also sometimes called *exact tests*, due to their statistical property of guaranteeing that the null model will not test as “significant” more than the alpha level of the test (see “[Statistical Significance and p-Values](#)” on page 103).

In a bootstrap permutation test, the draws outlined in steps 2 and 3 of the random permutation test are made *with replacement* instead of without replacement. In this way the resampling procedure models not just the random element in the assignment of treatment to subject but also the random element in the selection of subjects from a population. Both procedures are encountered in statistics, and the distinction between them is somewhat convoluted and not of consequence in the practice of data science.

Permutation Tests: The Bottom Line for Data Science

Permutation tests are useful heuristic procedures for exploring the role of random variation. They are relatively easy to code, interpret, and explain, and they offer a useful detour around the formalism and “false determinism” of formula-based statistics, in which the precision of formula “answers” tends to imply unwarranted certainty.

One virtue of resampling, in contrast to formula approaches, is that it comes much closer to a one-size-fits-all approach to inference. Data can be numeric or binary. Sample sizes can be the same or different. Assumptions about normally distributed data are not needed.

Key Ideas

- In a permutation test, multiple samples are combined and then shuffled.
- The shuffled values are then divided into resamples, and the statistic of interest is calculated.

- This process is then repeated, and the resampled statistic is tabulated.
- Comparing the observed value of the statistic to the resampled distribution allows you to judge whether an observed difference between samples might occur by chance.

Further Reading

- *Randomization Tests*, 4th ed., by Eugene Edgington and Patrick Onghena (Chapman & Hall/CRC Press, 2007)—but don't get too drawn into the thicket of nonrandom sampling
- *Introductory Statistics and Analytics: A Resampling Perspective* by Peter Bruce (Wiley, 2014)

Statistical Significance and p-Values

Statistical significance is how statisticians measure whether an experiment (or even a study of existing data) yields a result more extreme than what chance might produce. If the result is beyond the realm of chance variation, it is said to be statistically significant.

Key Terms for Statistical Significance and p-Values

p-value

Given a chance model that embodies the null hypothesis, the p-value is the probability of obtaining results as unusual or extreme as the observed results.

Alpha

The probability threshold of “unusualness” that chance results must surpass for actual outcomes to be deemed statistically significant.

Type 1 error

Mistakenly concluding an effect is real (when it is due to chance).

Type 2 error

Mistakenly concluding an effect is due to chance (when it is real).

Consider in [Table 3-2](#) the results of the web test shown earlier.

Table 3-2. 2×2 table for ecommerce experiment results

Outcome	Price A	Price B
Conversion	200	182
No conversion	23,539	22,406

Price A converts almost 5% better than price B ($0.8425\% = 200/(23539+200)*100$, versus $0.8057\% = 182/(22406+182)*100$ —a difference of 0.0368 percentage points), big enough to be meaningful in a high-volume business. We have over 45,000 data points here, and it is tempting to consider this as “big data,” not requiring tests of statistical significance (needed mainly to account for sampling variability in small samples). However, the conversion rates are so low (less than 1%) that the actual meaningful values—the conversions—are only in the 100s, and the sample size needed is really determined by these conversions. We can test whether the difference in conversions between prices A and B is within the range of *chance variation*, using a resampling procedure. By chance variation, we mean the random variation produced by a probability model that embodies the null hypothesis that there is no difference between the rates (see “[The Null Hypothesis](#)” on page 94). The following permutation procedure asks, “If the two prices share the same conversion rate, could chance variation produce a difference as big as 5%?”

1. Put cards labeled 1 and 0 in a box: this represents the supposed shared conversion rate of 382 ones and 45,945 zeros = $0.008246 = 0.8246\%$.
2. Shuffle and draw out a resample of size 23,739 (same n as price A), and record how many 1s.
3. Record the number of 1s in the remaining 22,588 (same n as price B).
4. Record the difference in proportion of 1s.
5. Repeat steps 2–4.
6. How often was the difference ≥ 0.0368 ?

Reusing the function `perm_fun` defined in “Example: Web Stickiness” on page 98, we can create a histogram of randomly permuted differences in conversion rate in R:

```
obs_pct_diff <- 100 * (200 / 23739 - 182 / 22588)
conversion <- c(rep(0, 45945), rep(1, 382))
perm_diffs <- rep(0, 1000)
for (i in 1:1000) {
  perm_diffs[i] = 100 * perm_fun(conversion, 23739, 22588)
}
hist(perm_diffs, xlab='Conversion rate (percent)', main='')
abline(v=obs_pct_diff)
```

The corresponding *Python* code is:

```
obs_pct_diff = 100 * (200 / 23739 - 182 / 22588)
print(f'Observed difference: {obs_pct_diff:.4f}%')
conversion = [0] * 45945
conversion.extend([1] * 382)
conversion = pd.Series(conversion)

perm_diffs = [100 * perm_fun(conversion, 23739, 22588)
               for _ in range(1000)]

fig, ax = plt.subplots(figsize=(5, 5))
ax.hist(perm_diffs, bins=11, rwidth=0.9)
ax.axvline(x=obs_pct_diff, color='black', lw=2)
ax.text(0.06, 200, 'Observed\ndifference', bbox={'facecolor':'white'})
ax.set_xlabel('Conversion rate (percent)')
ax.set_ylabel('Frequency')
```

See the histogram of 1,000 resampled results in Figure 3-5: as it happens, in this case the observed difference of 0.0368% is well within the range of chance variation.

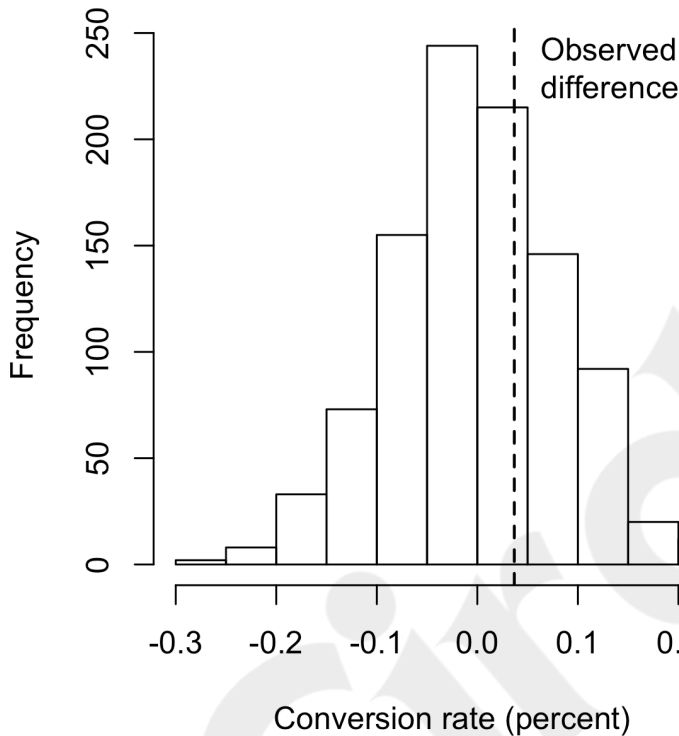


Figure 3-5. Frequency distribution for the difference in conversion rates between prices A and B

p-Value

Simply looking at the graph is not a very precise way to measure statistical significance, so of more interest is the *p-value*. This is the frequency with which the chance model produces a result more extreme than the observed result. We can estimate a *p-value* from our permutation test by taking the proportion of times that the permutation test produces a difference equal to or greater than the observed difference:

```
mean(perm_diffs > obs_pct_diff)
[1] 0.308

np.mean([diff > obs_pct_diff for diff in perm_diffs])
```

Here, both *R* and *Python* use the fact that true is interpreted as 1 and false as 0.

The *p-value* is 0.308, which means that we would expect to achieve a result as extreme as this, or a more extreme result, by random chance over 30% of the time.

In this case, we didn't need to use a permutation test to get a p-value. Since we have a binomial distribution, we can approximate the p-value. In R code, we do this using the function `prop.test`:

```
> prop.test(x=c(200, 182), n=c(23739, 22588), alternative='greater')

      2-sample test for equality of proportions with continuity correction

data:  c(200, 182) out of c(23739, 22588)
X-squared = 0.14893, df = 1, p-value = 0.3498
alternative hypothesis: greater
95 percent confidence interval:
 -0.001057439  1.000000000
sample estimates:
   prop 1      prop 2 
0.008424955 0.008057376
```

The argument `x` is the number of successes for each group, and the argument `n` is the number of trials.

The method `scipy.stats.chi2_contingency` takes the values as shown in Table 3-2:

```
survivors = np.array([[200, 23739 - 200], [182, 22588 - 182]])
chi2, p_value, df, _ = stats.chi2_contingency(survivors)

print(f'p-value for single sided test: {p_value / 2:.4f}')
```

The normal approximation yields a p-value of 0.3498, which is close to the p-value obtained from the permutation test.

Alpha

Statisticians frown on the practice of leaving it to the researcher's discretion to determine whether a result is “too unusual” to happen by chance. Rather, a threshold is specified in advance, as in “more extreme than 5% of the chance (null hypothesis) results”; this threshold is known as *alpha*. Typical alpha levels are 5% and 1%. Any chosen level is an arbitrary decision—there is nothing about the process that will guarantee correct decisions $x\%$ of the time. This is because the probability question being answered is *not* “What is the probability that this happened by chance?” but rather “Given a chance model, what is the probability of a result this extreme?” We then deduce backward about the appropriateness of the chance model, but that judgment does not carry a probability. This point has been the subject of much confusion.

p-value controversy

Considerable controversy has surrounded the use of the p-value in recent years. One psychology journal has gone so far as to “ban” the use of p-values in submitted papers on the grounds that publication decisions based solely on the p-value were resulting in the publication of poor research. Too many researchers, only dimly aware of what a

p-value really means, root around in the data, and among different possible hypotheses to test, until they find a combination that yields a significant p-value and, hence, a paper suitable for publication.

The real problem is that people want more meaning from the p-value than it contains. Here's what we would *like* the p-value to convey:

The probability that the result is due to chance.

We hope for a low value, so we can conclude that we've proved something. This is how many journal editors were interpreting the p-value. But here's what the p-value *actually* represents:

The probability that, *given a chance model*, results as extreme as the observed results could occur.

The difference is subtle but real. A significant p-value does not carry you quite as far along the road to “proof” as it seems to promise. The logical foundation for the conclusion “statistically significant” is somewhat weaker when the real meaning of the p-value is understood.

In March 2016, the American Statistical Association, after much internal deliberation, revealed the extent of misunderstanding about p-values when it issued a cautionary statement regarding their use. The **ASA statement** stressed six principles for researchers and journal editors:

1. P-values can indicate how incompatible the data are with a specified statistical model.
2. P-values do not measure the probability that the studied hypothesis is true, or the probability that the data were produced by random chance alone.
3. Scientific conclusions and business or policy decisions should not be based only on whether a p-value passes a specific threshold.
4. Proper inference requires full reporting and transparency.
5. A p-value, or statistical significance, does not measure the size of an effect or the importance of a result.
6. By itself, a p-value does not provide a good measure of evidence regarding a model or hypothesis.

Practical significance

Even if a result is statistically significant, that does not mean it has practical significance. A small difference that has no practical meaning can be statistically significant if it arose from large enough samples. Large samples ensure that small, non-meaningful effects can nonetheless be big enough to rule out chance as an explanation. Ruling out chance does not magically render important a result that is, in its essence, unimportant.

Type 1 and Type 2 Errors

In assessing statistical significance, two types of error are possible:

- A Type 1 error, in which you mistakenly conclude an effect is real, when it is really just due to chance
- A Type 2 error, in which you mistakenly conclude that an effect is not real (i.e., due to chance), when it actually is real

Actually, a Type 2 error is not so much an error as a judgment that the sample size is too small to detect the effect. When a p-value falls short of statistical significance (e.g., it exceeds 5%), what we are really saying is “effect not proven.” It could be that a larger sample would yield a smaller p-value.

The basic function of significance tests (also called *hypothesis tests*) is to protect against being fooled by random chance; thus they are typically structured to minimize Type 1 errors.

Data Science and p-Values

The work that data scientists do is typically not destined for publication in scientific journals, so the debate over the value of a p-value is somewhat academic. For a data scientist, a p-value is a useful metric in situations where you want to know whether a model result that appears interesting and useful is within the range of normal chance variability. As a decision tool in an experiment, a p-value should not be considered controlling, but merely another point of information bearing on a decision. For example, p-values are sometimes used as intermediate inputs in some statistical or machine learning models—a feature might be included in or excluded from a model depending on its p-value.

Key Ideas

- Significance tests are used to determine whether an observed effect is within the range of chance variation for a null hypothesis model.
- The p-value is the probability that results as extreme as the observed results might occur, given a null hypothesis model.
- The alpha value is the threshold of “unusualness” in a null hypothesis chance model.
- Significance testing has been much more relevant for formal reporting of research than for data science (but has been fading recently, even for the former).

Further Reading

- Stephen Stigler, “Fisher and the 5% Level,” *Chance* 21, no. 4 (2008): 12. This article is a short commentary on Ronald Fisher’s 1925 book *Statistical Methods for Research Workers* (Oliver & Boyd), and on Fisher’s emphasis on the 5% level of significance.
- See also “[Hypothesis Tests](#)” on page 93 and the further reading mentioned there.

t-Tests

There are numerous types of significance tests, depending on whether the data comprises count data or measured data, how many samples there are, and what’s being measured. A very common one is the *t-test*, named after Student’s *t*-distribution, originally developed by W. S. Gosset to approximate the distribution of a single sample mean (see “[Student’s t-Distribution](#)” on page 75).

Key Terms for t-Tests

Test statistic

A metric for the difference or effect of interest.

t-statistic

A standardized version of common test statistics such as means.

t-distribution

A reference distribution (in this case derived from the null hypothesis), to which the observed *t*-statistic can be compared.

All significance tests require that you specify a *test statistic* to measure the effect you are interested in and help you determine whether that observed effect lies within the range of normal chance variation. In a resampling test (see the discussion of permutation in “[Permutation Test](#)” on page 97), the scale of the data does not matter. You create the reference (null hypothesis) distribution from the data itself and use the test statistic as is.

In the 1920s and 1930s, when statistical hypothesis testing was being developed, it was not feasible to randomly shuffle data thousands of times to do a resampling test. Statisticians found that a good approximation to the permutation (shuffled) distribution was the t-test, based on Gosset’s t-distribution. It is used for the very common two-sample comparison—A/B test—in which the data is numeric. But in order for the t-distribution to be used without regard to scale, a standardized form of the test statistic must be used.

A classic statistics text would at this stage show various formulas that incorporate Gosset’s distribution and demonstrate how to standardize your data to compare it to the standard t-distribution. These formulas are not shown here because all statistical software, as well as *R* and *Python*, includes commands that embody the formula. In *R*, the function is `t.test`:

```
> t.test(Time ~ Page, data=session_times, alternative='less')

Welch Two Sample t-test

data: Time by Page
t = -1.0983, df = 27.693, p-value = 0.1408
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
 -Inf 19.59674
sample estimates:
mean in group Page A mean in group Page B
      126.3333          162.0000
```

The function `scipy.stats.ttest_ind` can be used in *Python*:

```
res = stats.ttest_ind(session_times[session_times.Page == 'Page A'].Time,
                      session_times[session_times.Page == 'Page B'].Time,
                      equal_var=False)
print(f'p-value for single sided test: {res.pvalue / 2:.4f}')
```

The alternative hypothesis is that the session time mean for page A is less than that for page B. The p-value of 0.1408 is fairly close to the permutation test p-values of 0.121 and 0.126 (see “[Example: Web Stickiness](#)” on page 98).

In a resampling mode, we structure the solution to reflect the observed data and the hypothesis to be tested, not worrying about whether the data is numeric or binary, whether or not sample sizes are balanced, sample variances, or a variety of other factors. In the formula world, many variations present themselves, and they can be

bewildering. Statisticians need to navigate that world and learn its map, but data scientists do not—they are typically not in the business of sweating the details of hypothesis tests and confidence intervals the way a researcher preparing a paper for presentation might.

Key Ideas

- Before the advent of computers, resampling tests were not practical, and statisticians used standard reference distributions.
- A test statistic could then be standardized and compared to the reference distribution.
- One such widely used standardized statistic is the t-statistic.

Further Reading

- Any introductory statistics text will have illustrations of the t-statistic and its uses; two good ones are *Statistics*, 4th ed., by David Freedman, Robert Pisani, and Roger Purves (W. W. Norton, 2007), and *The Basic Practice of Statistics*, 8th ed., by David S. Moore, William I. Notz, and Michael A. Fligner (W. H. Freeman, 2017).
- For a treatment of both the t-test and resampling procedures in parallel, see *Introductory Statistics and Analytics: A Resampling Perspective* by Peter Bruce (Wiley, 2014) or *Statistics: Unlocking the Power of Data*, 2nd ed., by Robin Lock and four other Lock family members (Wiley, 2016).

Multiple Testing

As we've mentioned previously, there is a saying in statistics: "Torture the data long enough, and it will confess." This means that if you look at the data through enough different perspectives and ask enough questions, you almost invariably will find a statistically significant effect.

For example, if you have 20 predictor variables and one outcome variable, all *randomly* generated, the odds are pretty good that at least one predictor will (falsely) turn out to be statistically significant if you do a series of 20 significance tests at the $\alpha = 0.05$ level. As previously discussed, this is called a *Type 1 error*. You can calculate this probability by first finding the probability that all will *correctly* test nonsignificant at the 0.05 level. The probability that *one* will correctly test nonsignificant is 0.95, so the probability that all 20 will correctly test nonsignificant is $0.95 \times 0.95 \times 0.95 \dots$, or

$0.95^{20} = 0.36$.¹ The probability that at least one predictor will (falsely) test significant is the flip side of this probability, or $1 - (\text{probability that all will be nonsignificant}) = 0.64$. This is known as *alpha inflation*.

This issue is related to the problem of overfitting in data mining, or “fitting the model to the noise.” The more variables you add, or the more models you run, the greater the probability that something will emerge as “significant” just by chance.

Key Terms for Multiple Testing

Type 1 error

Mistakenly concluding that an effect is statistically significant.

False discovery rate

Across multiple tests, the rate of making a Type 1 error.

Alpha inflation

The multiple testing phenomenon, in which *alpha*, the probability of making a Type 1 error, increases as you conduct more tests.

Adjustment of *p*-values

Accounting for doing multiple tests on the same data.

Overfitting

Fitting the noise.

In supervised learning tasks, a holdout set where models are assessed on data that the model has not seen before mitigates this risk. In statistical and machine learning tasks not involving a labeled holdout set, the risk of reaching conclusions based on statistical noise persists.

In statistics, there are some procedures intended to deal with this problem in very specific circumstances. For example, if you are comparing results across multiple treatment groups, you might ask multiple questions. So, for treatments A–C, you might ask:

- Is A different from B?
- Is B different from C?
- Is A different from C?

¹ The multiplication rule states that the probability of n independent events all happening is the product of the individual probabilities. For example, if you and I each flip a coin once, the probability that your coin and my coin will both land heads is $0.5 \times 0.5 = 0.25$.

Or, in a clinical trial, you might want to look at results from a therapy at multiple stages. In each case, you are asking multiple questions, and with each question, you are increasing the chance of being fooled by chance. Adjustment procedures in statistics can compensate for this by setting the bar for statistical significance more stringently than it would be set for a single hypothesis test. These adjustment procedures typically involve “dividing up the alpha” according to the number of tests. This results in a smaller alpha (i.e., a more stringent bar for statistical significance) for each test. One such procedure, the Bonferroni adjustment, simply divides the alpha by the number of comparisons. Another, used in comparing multiple group means, is Tukey’s “honest significant difference,” or *Tukey’s HSD*. This test applies to the maximum difference among group means, comparing it to a benchmark based on the *t-distribution* (roughly equivalent to shuffling all the values together, dealing out resampled groups of the same sizes as the original groups, and finding the maximum difference among the resampled group means).

However, the problem of multiple comparisons goes beyond these highly structured cases and is related to the phenomenon of repeated data “dredging” that gives rise to the saying about torturing the data. Put another way, given sufficiently complex data, if you haven’t found something interesting, you simply haven’t looked long and hard enough. More data is available now than ever before, and the number of journal articles published nearly doubled between 2002 and 2010. This gives rise to lots of opportunities to find something interesting in the data, including multiplicity issues such as:

- Checking for multiple pairwise differences across groups
- Looking at multiple subgroup results (“we found no significant treatment effect overall, but we did find an effect for unmarried women younger than 30”)
- Trying lots of statistical models
- Including lots of variables in models
- Asking a number of different questions (i.e., different possible outcomes)



False Discovery Rate

The term *false discovery rate* was originally used to describe the rate at which a given set of hypothesis tests would falsely identify a significant effect. It became particularly useful with the advent of genomic research, in which massive numbers of statistical tests might be conducted as part of a gene sequencing project. In these cases, the term applies to the testing protocol, and a single false “discovery” refers to the outcome of a hypothesis test (e.g., between two samples). Researchers sought to set the parameters of the testing process to control the false discovery rate at a specified level. The term has also been used for classification in data mining; it is the misclassification rate within the class 1 predictions. Or, put another way, it is the probability that a “discovery” (labeling a record as a “1”) is false. Here we typically are dealing with the case where 0s are abundant and 1s are interesting and rare (see [Chapter 5](#) and [“The Rare Class Problem” on page 223](#)).

For a variety of reasons, including especially this general issue of “multiplicity,” more research does not necessarily mean better research. For example, the pharmaceutical company Bayer found in 2011 that when it tried to replicate 67 scientific studies, it could fully replicate only 14 of them. Nearly two-thirds could not be replicated at all.

In any case, the adjustment procedures for highly defined and structured statistical tests are too specific and inflexible to be of general use to data scientists. The bottom line for data scientists on multiplicity is:

- For predictive modeling, the risk of getting an illusory model whose apparent efficacy is largely a product of random chance is mitigated by cross-validation (see [“Cross-Validation” on page 155](#)) and use of a holdout sample.
- For other procedures without a labeled holdout set to check the model, you must rely on:
 - Awareness that the more you query and manipulate the data, the greater the role that chance might play.
 - Resampling and simulation heuristics to provide random chance benchmarks against which observed results can be compared.

Key Ideas

- Multiplicity in a research study or data mining project (multiple comparisons, many variables, many models, etc.) increases the risk of concluding that something is significant just by chance.
- For situations involving multiple statistical comparisons (i.e., multiple tests of significance), there are statistical adjustment procedures.
- In a data mining situation, use of a holdout sample with labeled outcome variables can help avoid misleading results.

Further Reading

- For a short exposition of one procedure (Dunnett's test) to adjust for multiple comparisons, see David Lane's [online statistics text](#).
- Megan Goldman offers a [slightly longer treatment of the Bonferroni adjustment procedure](#).
- For an in-depth treatment of more flexible statistical procedures for adjusting p-values, see *Resampling-Based Multiple Testing* by Peter Westfall and Stanley Young (Wiley, 1993).
- For a discussion of data partitioning and the use of holdout samples in predictive modeling, see Chapter 2 of *Data Mining for Business Analytics*, by Galit Shmueli, Peter Bruce, Nitin Patel, Peter Gedeck, Inbal Yahav, and Kenneth Lichtendahl (Wiley, 2007–2020, with editions for *R*, *Python*, Excel, and JMP).

Degrees of Freedom

In the documentation and settings for many statistical tests and probability distributions, you will see a reference to “degrees of freedom.” The concept is applied to statistics calculated from sample data, and refers to the number of values free to vary. For example, if you know the mean for a sample of 10 values, there are 9 degrees of freedom (once you know 9 of the sample values, the 10th can be calculated and is not free to vary). The degrees of freedom parameter, as applied to many probability distributions, affects the shape of the distribution.

The number of degrees of freedom is an input to many statistical tests. For example, degrees of freedom is the name given to the $n - 1$ denominator seen in the calculations for variance and standard deviation. Why does it matter? When you use a sample to estimate the variance for a population, you will end up with an estimate that is

slightly biased downward if you use n in the denominator. If you use $n - 1$ in the denominator, the estimate will be free of that bias.

Key Terms for Degrees of Freedom

***n* or sample size**

The number of observations (also called *rows* or *records*) in the data.

d.f.

Degrees of freedom.

A large share of a traditional statistics course or text is consumed by various standard tests of hypotheses (t-test, F-test, etc.). When sample statistics are standardized for use in traditional statistical formulas, degrees of freedom is part of the standardization calculation to ensure that your standardized data matches the appropriate reference distribution (t-distribution, F-distribution, etc.).

Is it important for data science? Not really, at least in the context of significance testing. For one thing, formal statistical tests are used only sparingly in data science. For another, the data size is usually large enough that it rarely makes a real difference for a data scientist whether, for example, the denominator has n or $n - 1$. (As n gets large, the bias that would come from using n in the denominator disappears.)

There is one context, though, in which it is relevant: the use of factored variables in regression (including logistic regression). Some regression algorithms choke if exactly redundant predictor variables are present. This most commonly occurs when factoring categorical variables into binary indicators (dummies). Consider the variable “day of week.” Although there are seven days of the week, there are only six degrees of freedom in specifying day of week. For example, once you know that day of week is not Monday through Saturday, you know it must be Sunday. Inclusion of the Mon–Sat indicators thus means that *also* including Sunday would cause the regression to fail, due to a *multicollinearity* error.

Key Ideas

- The number of degrees of freedom (d.f.) forms part of the calculation to standardize test statistics so they can be compared to reference distributions (t-distribution, F-distribution, etc.).
- The concept of degrees of freedom lies behind the factoring of categorical variables into $n - 1$ indicator or dummy variables when doing a regression (to avoid multicollinearity).